

JEE MAIN REVISION MASTER: IONIC EQUILIBRIUM

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 140 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	pH Calculation of Acids, Bases & Salts	100 PYQs	pH & Hydrolysis
Very High	Solubility Product (Ksp) & Common Ion Effect	40 PYQs	Solubility

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
pH Calculation of Acids, Bases & Salts	$\text{pH} = \text{p}K_a + \log \frac{[\text{conjugate base}]}{[\text{acid}]}, \quad \text{Salt of WA-SB: } \text{pH} = 7 + \frac{1}{2} \text{p}K_a + \frac{1}{2} \log c$
Solubility Product (Ksp) & Common Ion Effect	$A_x B_y \rightleftharpoons x A^{y+} + y B^{x-} \implies K_{sp} = x^x y^y s^{x+y}, \quad Q_{sp} > K_{sp} \implies \text{Precipitation occurs}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 pH Calculation of Acids, Bases & Salts (Very High Yield — 100 PYQs)

Core Principle: Self-ionization of water ($K_w = 10^{-14}$ at 25°C), Ostwald dilution law, buffer solutions, and salt hydrolysis formulas.

■ **Caution / JEE Pitfall:** pH of 10^{-8} M HCl is NOT 8 (cannot be basic!); including water's auto-ionization $[H^+] = 10^{-8} + 10^{-7}$ implies $\text{pH} \approx 6.79$.

■ **Speed Trick:** Buffer capacity is maximum when $[\text{salt}] = [\text{acid}] \implies \text{pH} = \text{p}K_a$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Given is a concentrated solution of a weak electrolyte $A_x B_y$ of concentration ' c ' and dissociation constant ' K '. The degree of ..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "20 mL of a solution of acetic acid required 28.4 mL of 0.1 M NaOH for its neutralization. A solution (X) was prepared by mixing 20 mL of the..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Consider a weak base 'B' of $\text{p}K(b)=5.699$. ' x ' mL of 0.02 M HCl and ' y ' mL of 0.02 M weak base 'B' are mixed to make 100 mL of a..." [Ans: N/A]

3.2 Solubility Product (K_{sp}) & Common Ion Effect (Very High Yield — 40 PYQs)

Core Principle: Calculation of molar solubility s from K_{sp} , ionic product comparison, and suppression of solubility via common ion effect.

■ **Caution / JEE Pitfall:** For Ag_2CrO_4 (A_2B type): $K_{sp} = (2s)^2(s) = 4s^3 \implies s = \sqrt[3]{K_{sp}/4}$, NOT s^2 !

■ **Speed Trick:** Common ion effect ALWAYS suppresses the solubility of a sparingly soluble salt (e.g. AgCl is less soluble in NaCl than pure water).

Representative Source PYQs:

- **[2026 Morning Shift] Q#2 (MCQ):** " $\text{M}_{(3)}\text{A}_{(2)}$ is a sparingly soluble salt of molar mass $y\text{g mol}^{-1}$ and solubility $x\text{g L}^{-1}$. The ratio of the molar concentration of the ..." **[Ans: A]**
- **[2026 Morning Shift] Q#4 (MCQ):** "The solubility product constants of $\text{Ag}_{(2)}\text{CrO}_{(4)}$ and AgBr are $32x$ and $4y$ respectively at 298 K. The value of $\frac{\text{molarity} \dots}{\text{molarity} \dots}$ " **[Ans: D]**
- **[2025 Morning Shift] Q#1 (MCQ):** "If equal volumes of $\text{AB}_{(2)}$ and XY (both are salts) aqueous solutions are mixed, which of the following combination will give a precipitate..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 21st January Evening Shift (Q#1)

The first and second ionization constants of H_2X are 2.5×10^{-8} and 1.0×10^{-13} respectively. The concentration of X^{2-} in 0.1M H_2X solution is $_____ \times 10^{-15}\text{M}$. (Nearest Integer)

Correct Answer: 100

Solution Analysis: For the diprotic acid H_2X (total concentration $C=0.1\text{M}$): $K_{a1}=2.5 \times 10^{-8}, K_{a2}=1.0 \times 10^{-13}$
 Find $[\text{H}^+]$ (second dissociation is negligible for $[\text{H}^+]$): $[\text{H}^+] \approx \sqrt{(K_{a1}C)} = \sqrt{(2.5 \times 10^{-8})(0.1)} = \sqrt{2.5 \times 10^{-9}} = 5.0 \times 10^{-5}\text{M}$
 Use diprotic distribution for $[\text{X}^{2-}]$: $[\text{X}^{2-}] = C \cdot \frac{K_{a1}K_{a2}}{[\text{H}^+]^2 + K_{a1} + K_{a1}K_{a2}/[\text{H}^+]}$
 Compute key terms: $\frac{K_{a1}K_{a2}}{[\text{H}^+]^2} = \frac{(2.5 \times 10^{-8})(1.0 \times 10^{-13})}{(5 \times 10^{-5})^2} = \frac{2.5 \times 10^{-21}}{2.5 \times 10^{-9}} = 10^{-12}$

Benchmark Problem 2: JEE Main 2026 (Online) 5th April Morning Shift (Q#2)

$\text{M}_{(3)}\text{A}_{(2)}$ is a sparingly soluble salt of molar mass $y\text{g mol}^{-1}$ and solubility $x\text{g L}^{-1}$. The ratio of the molar concentration of the anion (A^{3-}) to the solubility product of the salt is

- (A) $(1/54) \cdot \frac{x^4}{y^4}$ **[Correct]**
- (B) $\frac{x^5}{y^4}$
- (C) $108 \cdot \frac{x^5}{y^5}$
- (D) $(1/108) \frac{x^4}{y^4}$

Correct Answer: A

Solution Analysis: Given: Solubility of the salt in $\text{g L}^{-1} = x$ Molar mass of the salt in $\text{g mol}^{-1} = y$ The molar solubility S (in mol L^{-1}) is given by: $S = \frac{\text{Solubility in g L}^{-1}}{\text{Molar mass}} = (x/y)$ Now, let us write the dissociation equation for the sparingly soluble salt $\text{M}_{(3)}\text{A}_{(2)}$ in water. According to the stoichiometry, one mole of the salt produces three moles of the cation and two moles of the anion. $\text{M}_{(3)}\text{A}_{(2)} \rightleftharpoons 3\text{M}^{2+} + 2\text{A}^{3-}$ If S is the molar solubility of the salt, the equilibrium concentrations of the ions will be: Molar